

HANIF ARDIYANTA NUGRAHA

| hanifardiyanta11@gmail.com | linkedin.com/in/hanifardiyantanugraha | nugrahax.vercel.app

Yogyakarta, Indonesia

Software Engineer with experience in end-to-end software system development, from backend and frontend design to integration with IoT-based hardware. Possesses expertise in building web applications, real-time monitoring systems, and firmware for microcontrollers such as ESP32. Accustomed to developing solutions that not only function as software but are also capable of controlling and communicating directly with physical devices via network protocols. Oriented towards building efficient, scalable system architectures that deliver a real impact on users.

PENDIDIKAN

Universitas Gadjah Mada - Yogyakarta

Jul 2023 - Aug 2027 (Expected)

Bachelor of Electronics and Instrumentation

EXPERIENCE

Koding Next Yogyakarta - Yogyakarta

Aug 2025 - Present

Advanced Coding Teacher

- Taught over 70 students across 6 modules, including Python Arcade, Lua (Roblox Studio), Computer Vision (Object Detection), and IoT (Intruder Detection Systems).
- Appointed as Technical Judge for the International Environment Coding Competition (IECC) 2025 for the Central Java & Yogyakarta region, selecting top participants to advance to the national level.
- Delivered a tech talk at Yogyakarta GIK UGM 2025 to over 50 participants, discussing how hands-on coding practices can unlock digital potential for everyone.
- Mentored students from the conceptual design stage to real-time hardware and software debugging, ensuring optimal project quality.

Elins Research Club UGM - Yogyakarta

Apr 2025 - Present

Senior Member of WebDev Division

- Independently organized and delivered an Advanced JavaScript workshop for new members to elevate the organization's technical standards and web development competencies.
- Executed a Community Service program at MAN 2 Yogyakarta by developing a full-stack e-learning platform using JavaScript and Python, integrated with student-made IoT devices for real-time data visualization.
- Developed Weather Station software as part of a community-based project, featuring real-time data monitoring, historical logging, and automated notifications based on weather conditions.
- Developed a specialized Nutrition Calculator web application for the National Nutrition Agency (BGN) to support the national Free Nutritious Meal (MBG) initiative.
- Maintained and optimized erc.elins.id to showcase research portfolios, community service documentation, and technical projects to institutional partners.

Electronics and Instrumentation Student Association UGM - Yogyakarta

Feb 2025 - Feb 2026

Head of Academic and Advocacy Division

- Initiated impactful career development programs, including a socialization event with the Apple Developer Academy @ Infinite Learning Batam, connecting students with global industry opportunities.
- Improved academic outcomes by organizing tutorial classes named "ELINS Tutorial" for 100+ new students and creating a centralized digital bank for exam questions and course materials (Elinsklopedia).
- Initiated and led the EFO (Elins Foundation) scholarship program, successfully establishing partnerships with alumni and donors; responsibilities ranged from direct communication and intensive consultation with faculty advisors to designing a comprehensive selection scheme and managing all administrative processes and post-acceptance accountability reports (LPJ).
- Led and coordinated a 12-member division in executing strategic programs that directly impacted student welfare and alumni-campus connections.
- Developed student advocacy programs to assist with academic processes and convey student rights to the faculty, and created weekly technology-based academic quizzes on the association's social media.

Computer and Network Systems Laboratory UGM - Yogyakarta

Jul 2025 - Oct 2025

Software Developer

- Led the software architecture design and developed modular firmware (C++) for the ESP32 microcontroller as the core of an end-to-end IoT monitoring system.
- Developed a centralized data gateway using Python, FastAPI, and Docker to process and stream sensor data in real-time.
- Designed and implemented an interactive data visualization dashboard using Next.js.

Basic Electronics Laboratory UGM - Yogyakarta

Feb 2025 - Jul 2025

Laboratory Practicum Assistant

- Led and facilitated practicum sessions for approximately 30 undergraduate students with full responsibility for grading and learning outcomes.
- Taught comprehensive electronics skills, including schematic and PCB layout design using EAGLE, PCB fabrication (etching, drilling, soldering), and final testing and troubleshooting.
- Designed and evaluated case-study-based practicum exams to measure students' technical competencies and problem-solving abilities.
- Provided individual mentoring for one semester to selected students to deepen conceptual understanding and ensure project success.

Basic Programming Laboratory UGM - Yogyakarta

Aug 2025 - Dec 2025

Programming Practicum Assistant

- Compiled and prepared C++ programming learning modules from basic to intermediate levels in English as official practicum guidelines.
- Coordinated with lecturers and the laboratory team to ensure modules were systematic and aligned with the course's learning outcomes.
- Reviewed and refined materials to ensure clarity, technical consistency, and the academic quality of the modules.
- Supported the laboratory's administrative and technical needs to ensure smooth programming learning processes.

Electronics and Instrumentation UGM - Yogyakarta

Aug 2025 - Dec 2025

Programming Course Tutor

- Provided structured learning over one semester to more than 60 new students to build fundamental understanding in C++ programming.
- Guided students in understanding basic to advanced concepts such as control structures, functions, arrays, pointers, and the basics of object-oriented programming.
- Assisted in Midterm (UTS) and Final Exam (UAS) preparation through question review sessions, targeted exercises, and exam simulations.
- Actively answered technical questions both in class and through online discussion groups to ensure all students understood the material equally.
- Helped students develop problem-solving skills and algorithmic logic through practical approaches and case studies

PROJECT EXPERIENCE

MBG Smart Nutrition Calculator - Yogyakarta

Jul 2025 - Sep 2025

Software Developer

Next.js, Express.js, Prisma ORM, Gemini AI

- Developed a web application to calculate food nutritional values and generate nutrition labels automatically to support the national Free Nutritious Meal (MBG) program.
- Built the backend using Express.js and Prisma ORM for managing food ingredient data and nutritional composition.
- Integrated AI (Gemini) to estimate the nutritional content of new food ingredients not yet available in the database.
- Developed the user interface using Next.js to ensure a responsive and user-friendly experience.
- Optimized nutrition calculation accuracy through comprehensive data validation and system testing.

G-Connect Realtime Monitoring Weather Station

Lead Software Developer

Next.js, FastAPI, SQLite

- Led the development of an IoT-based real-time weather monitoring system integrated with ESP32 sensors to support agricultural decision-making.
- Designed and implemented a REST API using FastAPI to process and manage environmental data (temperature, humidity, soil moisture) in real-time.
- Developed an interactive dashboard using Next.js featuring historical data visualization and automated notifications based on weather conditions.
- Designed a lightweight and efficient SQLite database architecture for implementation in rural environments with limited resources.
- Collaborated with the hardware team on HTTP-based data communication integration to ensure stable data transmission.

GamaSense Monitoring Air Quality

Software Developer

Typescript, React + Vite, Node-RED, SQLite

- Developed an IoT-based air quality monitoring system that displays real-time gas sensor data through a web dashboard.
- Built a data processing pipeline using Node-RED to manage and distribute sensor data efficiently.
- Developed a responsive interface using React + Vite with TypeScript to improve code scalability and maintainability.
- Designed the database structure for recording air quality data and analyzing long-term trends.
- Contributed to improving system stability through modular testing and application performance optimization.

Web Elins Research Club UGM

Software Developer

Wordpress Elementor

- Developed and maintained the official club website (erc.elins.id), utilizing WordPress and Elementor to build a professional organizational profile web and information center for members and the public.
- Designed and led JavaScript training sessions for new members to accelerate technical skill development and integration into the development team.
- Collaborated within the development team to translate client needs into functional web solutions, taking primary responsibility for implementing specific features and ensuring code quality.

Automation AC Controller Mobile App

Mobile Developer

Kotlin, Jetpack Compose

- Developed an Android application to control IoT-based AC systems using Kotlin and Jetpack Compose.
- Integrated the application with microcontroller devices via the MQTT protocol for remote temperature and mode control.
- Designed an intuitive UI/UX featuring real-time device status monitoring.
- Implemented command synchronization mechanisms and connection validation to ensure stable communication between the application and the device.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, C++, Kotlin, SQL
- **Web & Backend Development:** React.js, Next.js, Node.js, Express.js, FastAPI, Flask
- **Mobile & IoT Systems:** Android (Kotlin, Jetpack Compose), ESP32, MQTT, HTTP, Raspberry Pi
- **Database & Tools:** MySQL, SQLite, Prisma ORM, Git, Docker
- **GamaSense Air Quality Monitoring** 